Week 9: Backtesting, Risk Analysis & Taker Strategies

Assembling the full system – Milestone 3 (Coincall guest session)

North Carolina State University | Summer 2026

NC STATE UNIVERSITY

Industry Mentor: Dendi Suhubdy (Bitwyre) | Guest: Fenni Kang (Coincall)

Learning Objectives

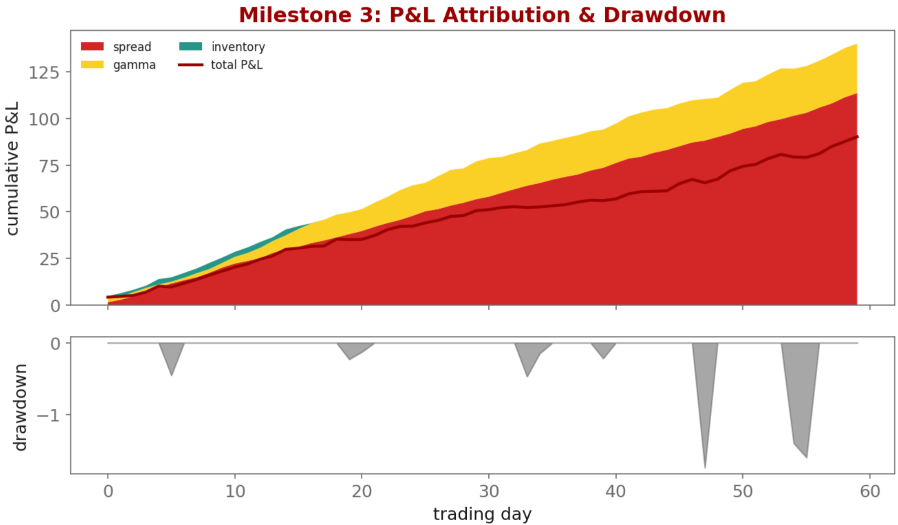
- Build an event-driven options backtest
- Produce a full P&L attribution and risk metrics
- Understand taker strategies and order-flow toxicity

- Event-driven simulation with deterministic replay
- Fill simulator; state of inventory and Greeks
- Pitfalls: look-ahead, unrealistic fills, expiry handling
- Sharpe with block-bootstrap confidence intervals

Lecture 2 (Guest: Fenni Kang, Coincall): Taker Strategies

- When and why takers cross the spread
- Signals behind aggressive flow: momentum, liquidations, news
- What taker behaviour means for the maker's toxicity

Illustration



Derivation & Rationale

$$\widehat{SR} = \frac{\bar{r}}{s_r}\sqrt{252}, \quad \text{CI via block bootstrap on } \{r_t\}$$

Why: Daily Sharpe is annualized by $\sqrt{252}$, but a point estimate over a few months is noisy and serially correlated. We resample BLOCKS of returns (preserving autocorrelation) to get an honest confidence interval – a high Sharpe with a wide CI is not a real edge.

Getting the Data from Coincall

- Full options L2 + trades + perp + funding for 3 months:
  - `coincall_btc_options_l2_2025Q4.parquet / ..._trades_...`
  - `coincall_eth_options_l2_2025Q4.parquet / ..._trades_...`
  - `coincall_btc_eth_perp_l2_2025Q4.parquet ; coincall_funding_rates_...`
- Klines for regime context: GET /open/option/market/kline/history/v1/{name}?period=h1
- Trade tape carries mark IV at trade -> calibrate the fill simulator

Cross-check exact paths at https://docs.coincall.com/

Code — week9_attribution.py

```
import numpy as np
def attribute(spread, inventory, gamma, hedging):
    total = spread + inventory + gamma + hedging
    assert abs(total.sum() - (spread.sum()+inventory.sum()
               +gamma.sum()+hedging.sum())) < 1e-6 # must sum to total
    return dict(spread=spread, inventory=inventory, gamma=gamma,
                hedging=hedging, total=total)

def bootstrap_sharpe(r, block=5, n=2000, seed=0):
    rng = np.random.default_rng(seed); T = len(r); out = []
    for _ in range(n):
        idx = (rng.integers(0, T-block, T//block)[:,None] + np.arange(block)).ravel()
        s = r[idx]; out.append(s.mean()/s.std()*np.sqrt(252))
    return np.percentile(out, [2.5, 50, 97.5]) # CI for the Sharpe
```

Glossary — Key Terms

Event-driven backtest a simulation that replays timestamped events in order

Deterministic replay bitwise-identical results on re-run

Look-ahead bias using information not available at decision time

Sharpe ratio risk-adjusted return: annualized mean over standard deviation

Block bootstrap resampling blocks of returns for a CI that keeps autocorrelation

Maximum drawdown the largest peak-to-trough decline in equity

Taker strategy aggressive, spread-crossing execution

Reading

- Cartea et al., Ch. 10-11; Bailey-Lopez de Prado (2014); Lopez de Prado (2018)

References

- Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading, Ch. 10-11.
- Bailey, D. H., & Lopez de Prado, M. (2014). The deflated Sharpe ratio. Journal of Portfolio Management, 40(5), 94-107.
- Lopez de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.
- Coincall API documentation. https://docs.coincall.com/